

Heart failure with preserved ejection fraction in adults with congenital heart disease

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Abstract

Background and Aims

Heart failure with preserved ejection fraction (HFpEF) is increasingly recognized in the general population, but the prevalence among adults with congenital heart disease (ACHD) is not well described. The study aimed to determine the prevalence, characteristics, and hospitalization burden of HFpEF among ACHD patients.

Methods

Using the electronic medical files, a retrospective, multicentre cohort study was conducted including 4507 ACHD patients with biventricular heart physiology and systemic left ventricle followed at the ACHD centres in Eastern Denmark. Heart failure with preserved ejection fraction was defined as preserved left ventricular ejection fraction (LVEF $\geq 50\%$) with diuretics use, combined with elevated N-terminal pro-B-type natriuretic peptide and/or echocardiographic signs of HFpEF.

Results

Among 4507 ACHD patients, 86% had a preserved LVEF. Heart failure with preserved ejection fraction was identified in 13%, with high prevalence across all ACHD severities (13%, 12%, and 18% in mild, moderate, and severe). Compared with patients with normal left ventricular function, HFpEF patients were older {median age: 60 [interquartile range (IQR) 47–70] vs 41 [IQR 29–56] years, $P < .001$ }, had more comorbidities {age-adjusted odds ratio for diabetes: 15.60 [95% confidence interval (CI) 12.39–19.70]; obesity: 1.48 [95% CI 1.17–1.87], and atrial fibrillation 4.58 [95% CI 2.22–9.88]}, and experienced almost five-fold higher rate of cardiovascular hospitalization [incidence rate ratio 4.6 (95% CI 4.2–5.1), $P < .0001$]. Despite this, only 4.0% had a heart failure diagnosis, and 5.8% were treated with a sodium-glucose cotransporter 2 (SGLT2) inhibitor.

Conclusions

Heart failure with preserved ejection fraction is highly prevalent among ACHD patients, regardless of ACHD severity. It is associated with an increased burden of cardiovascular risk factors and hospitalizations. Despite this, only a minority received SGLT2 inhibitors. These findings emphasize the need for increased awareness of HFpEF in the ACHD population.

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Structured Graphical Abstract

Key Question

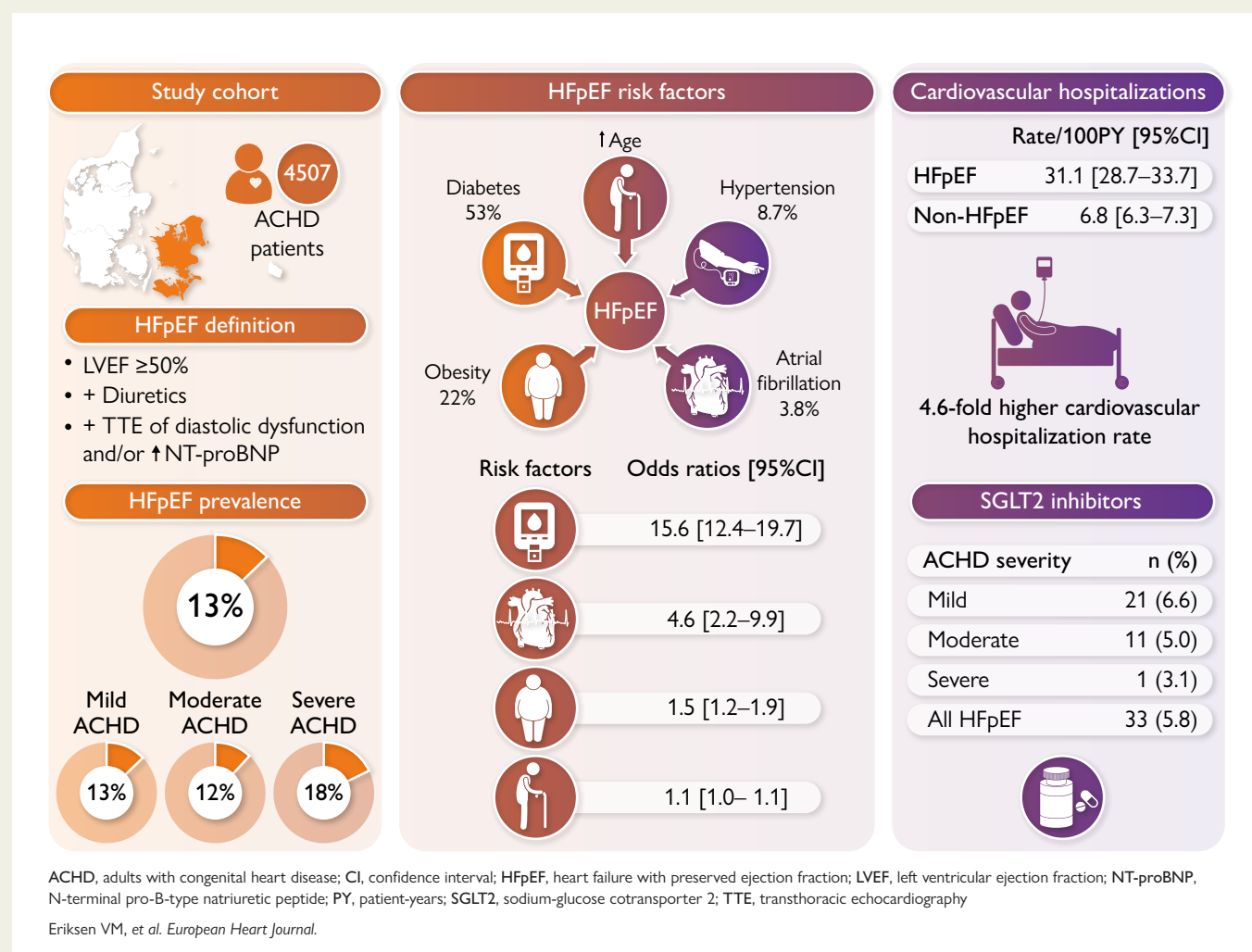
What is the prevalence of heart failure with preserved ejection fraction (HFpEF) in adults with congenital heart disease (ACHD)? What are the associations with comorbidities, hospitalizations, and treatment with sodium-glucose cotransporter 2 inhibitors (SGLT2i)?

Key Finding

HFpEF was identified in 13% of all ACHD patients, with a high prevalence across all ACHD severities. HFpEF was associated with older age, higher burden of comorbidities and hospital admissions. However, few had a recorded heart failure diagnosis or received SGLT2i.

Take Home Message

HFpEF is underdiagnosed in patients with ACHD. Clinical awareness is essential for effective management across all ACHD severity levels.



Keywords

Adult congenital heart disease (ACHD) • Heart failure with preserved ejection fraction (HFpEF) • Diastolic dysfunction • Hospitalization • SGLT2 inhibitor

Introduction

Heart failure (HF) is one of the leading causes of morbidity and mortality globally. The prevalence of HF with preserved ejection fraction (HFpEF) is increasing and is associated with high rates

of hospitalization and mortality, accounting for approximately half of all HF cases.^{1,2} Despite growing recognition of HFpEF in the general population, its prevalence and clinical significance in adult patients with congenital heart disease (ACHD) remain poorly described.

Heart failure with preserved ejection fraction is likely more common in ACHD patients than in the general population due to the presence of structural defects, early-life cyanosis, scar tissue from prior cardiac surgeries or interventions, and frequent conduction abnormalities or arrhythmias.^{3–5} Additionally, many ACHD patients present with risk factors such as systemic hypertension and obesity at a young age, and due to improved survival, the increasing age of the population may further raise their risk of developing HFpEF.^{5–10}

Diagnosing HFpEF in the general population is challenging, as multiple other conditions must be ruled out; it may be even more difficult in ACHD patients. Symptoms such as exercise intolerance and dyspnoea are often attributed to the underlying congenital heart condition, potentially leading to underdiagnosis of HFpEF in this group.^{4,11}

With recently published evidence demonstrating the clinical benefits of sodium-glucose cotransporter 2 (SGLT2) inhibitors^{12,13} in reducing hospitalization in HFpEF, and considering the rising HF hospitalization rates among ACHD patients, increased awareness of HFpEF in this population is of growing importance.

Therefore, the aim of this study was to investigate the prevalence of HFpEF among patients with ACHD. Furthermore, we aimed to characterize the HFpEF population and examine associated hospitalizations and medication use.

Methods

Design

This was a retrospective observational, multicentre cohort study.

Study population

The study included all Danish patients aged ≥ 18 years with congenital heart disease (ICD-10: DQ20–DQ26), who were followed at one of the four university hospitals responsible for ACHD care in Eastern Denmark (Rigshospitalet, Herlev-Gentofte Hospital, Bispebjerg-Frederiksberg Hospital, and Zealand University Hospital). All ACHD patients are managed exclusively within the public healthcare system, with echocardiography being routinely performed at each ACHD follow-up visit, while blood investigation being performed when indicated. All clinical information during the follow-up visit is systematically documented in electronic health records, from which study data were extracted. An ACHD specialist validated the ACHD diagnoses through review of medical records.

Patients were excluded if they had undergone heart transplantation, received resynchronization therapy, or had a syndrome without a congenital heart lesion. Individuals with a univentricular heart or a systemic right ventricle (RV) were excluded from the main analysis but were assessed separately in a subanalysis. Patients without a documented left ventricular ejection fraction (LVEF) between 1 September 2019 and 1 September 2024 were also excluded (see [Supplementary data online, Figure S1](#)). The most recent LVEF was categorized as preserved ($\geq 50\%$), mildly reduced ($40\%–49\%$), or reduced ($\leq 40\%$).¹⁴ The ACHD severity was categorized as mild, moderate, or severe, according to the 2020 ESC ACHD guidelines.¹⁵ A full list of CHD inclusion diagnoses and their respective severity classification is outlined in [Supplementary data online, Table S1](#).

Outcome

Patients were classified as having HFpEF if they met the HFpEF simplified diagnostic criteria outlined in 2021 ESC HF guidelines^{14,16}:

an LVEF $\geq 50\%$ in combination with diuretic treatment as a surrogate marker for symptoms and signs of HF. In addition, patients were required to have either an elevated N-terminal pro-B-type natriuretic peptide (NT-proBNP) [>365 pg/mL in patients with atrial fibrillation (AF) or >125 pg/mL in sinus rhythm (SR)] and/or one of the following echocardiographic parameters: (i) left atrial volume index (LAVI) > 40 mL/m² in the presence of AF, or LAVI >34 mL/m² in SR, (ii) E/e' ratio > 9 , (iii) tricuspid regurgitation velocity > 2.8 m/s, or (iv) increased left ventricular (LV) mass index, defined as ≥ 95 g/m² in females or ≥ 115 g/m² in males, or relative wall thickness > 0.42 .¹⁴

For the subanalysis of patients with a univentricular heart or a systemic RV, classification as HFpEF required preserved systemic ventricular ejection fraction, diuretic treatment, and elevated NT-proBNP levels.

The most recent echocardiographic examination performed during the ACHD clinic visits between 1 September 2019 and 1 September 2024, and related blood investigation, including NT-proBNP, usually taken with indications such as dyspnoea, were used to analyse the prevalence of HFpEF.

Cardiovascular hospitalizations

Hospital admissions were identified via unique contact identification number in the electronic health records. Cardiovascular hospitalization was defined as either admission to the department of cardiology or admission to the emergency department with relevant cardiovascular diagnoses such as HF, ischaemic heart disease, arrhythmias, endocarditis, aortic aneurysm or dissection, thromboembolism, and cardiac diagnostic or therapeutic procedure. Only admissions lasting for ≥ 24 h were included, with admissions for elective cardiac surgeries or procedures excluded. Hospitalization rate was calculated as number of hospitalizations per 100 patient-years (PY), where follow-up began on 1 September 2019 and continued until either death or end of study on 1 September 2024. As follow-up could begin prior to HFpEF diagnosis was established, there is a potential period where it is likely that patients had undiagnosed HFpEF, and hospitalizations were related to this. However, as this is an assumption, we also did a sensitivity analysis starting from the day of echocardiography (as 99% of HFpEF diagnoses were based on echocardiography).

Risk factors associated with heart failure with preserved ejection fraction

Comorbidities such as AF, systemic hypertension, chronic obstructive pulmonary disease, pulmonary arterial hypertension (PAH), chronic kidney disease (CKD), diabetes, obesity, and anaemia were identified using ICD-10 codes, medications prescribed at each ACHD follow-up visit or hospitalization, and/or blood test results, as detailed in [Supplementary data online, Table S2](#). As prescriptions are routinely renewed at each ACHD follow-up visit and during hospital discharge, any medications prescribed for these chronic conditions during the study period are considered as baseline comorbidity, regardless of the date of prescription. Obesity was defined as a body mass index ≥ 30 kg/m²,¹⁷ based on the most recently recorded height and weight, which is routinely registered at each echocardiography performance. Anaemia was defined as haemoglobin < 13 g/dL in men and < 12 g/dL in women, according to the World Health Organization criteria.¹⁸ For blood tests, the mean of the three most recent values was used when available; if fewer were available, the mean of the two most recent or the single most recent value was applied.

Statistical analysis

Continuous variables are reported as means with $\pm$ standard deviation (SD) or medians with interquartile range (IQR), and trends

across ACHD severity were assessed using linear regression. Categorical variables are presented as frequencies with percentages (%) and compared using the Cochran–Armitage trend test. Ordinal variables were compared using the Kruskal–Wallis test. Comparisons were made between patients meeting the HFpEF criteria and those with preserved LVEF not meeting the HFpEF criteria (hereafter referred to as normal LV function) using χ^2 tests for categorical variables and Wilcoxon rank-sum tests for continuous variables. Logistic regression model was employed to assess the associations between HFpEF and baseline comorbidities in the whole ACHD cohort, adjusted for age and ACHD severity. In the stratified analysis by ACHD severity, associations between HFpEF and baseline comorbidities were adjusted for age. Only diabetes, AF, obesity, and hypertension were tested for the association with HFpEF, based on their established clinical relevance in HFpEF and their sufficient prevalence in the study cohort to allow stable model estimation. Results are reported as odds ratios (ORs) with 95% confidence intervals (CIs). Hospitalization rates were analysed using Poisson regression with an offset of log-transformed follow-up time. Comparison of hospitalization rates between patients classified as HFpEF and patients with normal LV function was adjusted for age and ACHD severity. Results are reported as incidence rate ratios (IRRs) and model-based incidence rates/100 PY with 95% CIs. A two-sided *P*-value of $<.05$ was considered statistically significant. All analyses were performed in R (v4.4.3) using RStudio(v2024.09.1-394).

Results

A total of 4507 ACHD patients with a biventricular heart and a systemic left ventricle were identified. Of these, 86% ($n = 3896$) had a preserved LVEF. Nineteen per cent ($n = 732$) of patients with preserved LVEF were treated with diuretics. The distribution of ACHD severity among patients with preserved LVEF was 58%, 39%, and 3.5% for the mild, moderate, and severe ACHD, respectively. Additional demographic characteristics are presented in [Supplementary data online, Table S3](#).

Heart failure with preserved ejection fraction in adult patients with congenital heart disease

Thirteen per cent ($n = 572$) of patients in the biventricular cohort met the ESC HFpEF diagnostic criteria. The prevalence was 13% in mild, 12% in moderate, and 18% in severe ACHD. The majority of the patients (58%) who met the ESC HFpEF diagnostic criteria had more than one abnormal parameter for HFpEF ([Figure 1](#)), and increased LV mass index was the most common abnormal parameter (60.1%) ([Figure 2](#)). Adult patients with congenital heart disease meeting HFpEF criteria were older compared with the ACHD patients with normal LV function [median 60 years (IQR 47–70) vs 41 years (IQR 29–56), $P < .001$], but the age distribution varied by ACHD severity, with mild ACHD patients being older [median 64 years (IQR 55–73)] than those with moderate [54 years (IQR 35–66)] and severe ACHD [52 years (IQR 33–62)] (P for trend $< .0001$). [Supplementary data online, Figure S2](#) showed an increasing prevalence of HFpEF with increasing age.

Among the mild CHD patients fulfilling the ESC HFpEF diagnostic criteria, aortic valve disease was the most frequent diagnosis (76.4%), followed by atrial septal defect (11.6%) and pulmonary valve disease (7.2%). Among moderate CHD,

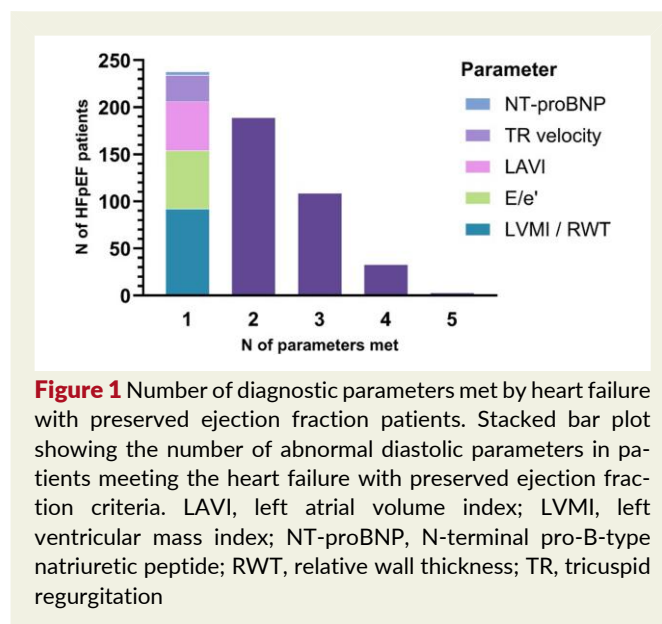


Figure 1 Number of diagnostic parameters met by heart failure with preserved ejection fraction patients. Stacked bar plot showing the number of abnormal diastolic parameters in patients meeting the heart failure with preserved ejection fraction criteria. LAVI, left atrial volume index; LVMI, left ventricular mass index; NT-proBNP, N-terminal pro-B-type natriuretic peptide; RWT, relative wall thickness; TR, tricuspid regurgitation

atrioventricular septal defect (58.6%), tetralogy of Fallot (TOF) (20.7%), and coarctation of the aorta (11.3%) were most common (see [Supplementary data online, Figure S3](#)). Among patients with severe CHD fulfilling the ESC HFpEF diagnostic criteria, Eisenmenger (46.9%), pulmonary atresia (18.8%), and double outlet RV (12.5%) predominated. Right ventricular function (TAPSE > 17 mm) was preserved in 67% ($n = 284$) of the HFpEF patients with available TAPSE measurements.

Heart failure with preserved ejection fraction and associated risk factors

After adjustment for age and ACHD severity, diabetes [OR 15.6 (12.39–19.70)] showed the strongest overall association with meeting HFpEF criteria. Atrial fibrillation [OR 4.58 (2.22–9.88)] and obesity [OR 1.48 (1.17–1.87)] were also significantly associated. Across ACHD severities, AF showed association with meeting HFpEF criteria in mild and moderate CHD and obesity only in mild CHD ([Figure 3](#)).

Cardiovascular hospitalizations

Patients meeting the HFpEF criteria had a near five-fold higher rate of cardiovascular hospitalization compared with those with normal LV function, after adjusting for age and ACHD severity [IRR: 4.6 (95% CI 4.2–5.1), $P < .0001$]. Crude hospitalization rates were 32.2/100 PY (95% CI 30.2–34.3) in those classified as HFpEF and 5.6/100 PY (95% CI 5.3–6.0) in patients with normal LV function. The increased cardiovascular hospitalization burden among those classified as HFpEF was consistent across the ACHD severities, with the highest hospitalization rate observed in severe ACHD [49.5/100 PY (95% CI 39.6–61.9)] ([Table 1](#)).

Sensitivity analysis with follow-up time starting from day of echocardiography showed that, during a median follow-up of 1.2 years (IQR 0.5–2.4), hospitalization rate for those classified as HFpEF was 24.7/100 PY (95% CI 21.5–28.4) and 5.5/100 PY (95% CI 4.9–6.1) for patients with normal LV function. This represented a 3.4-fold higher hospitalization rate for HFpEF

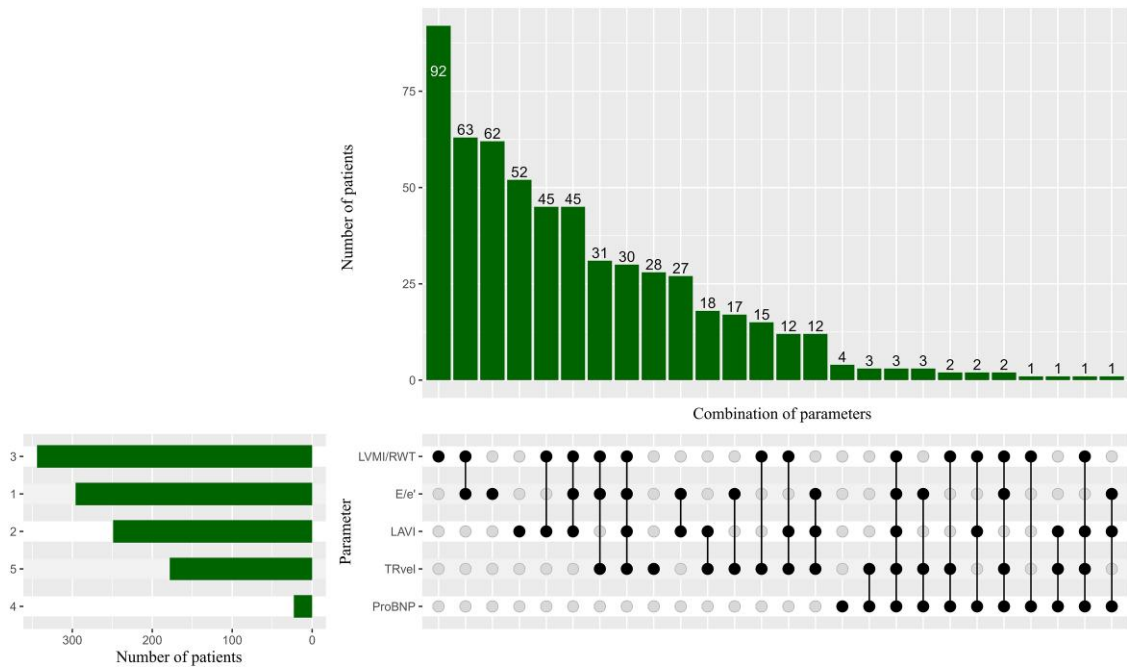


Figure 2 Distribution of combinations of diagnostic parameters met by heart failure with preserved ejection fraction patients. UpSet plot illustrating the combinations of diagnostic criteria fulfilled by the heart failure with preserved ejection fraction patients. Vertical bars represent the number of patients meeting each combination of parameters, and horizontal bars indicate the number of patients fulfilling each individual parameter. LAVI, left atrial volume index; LVMI, left ventricular mass index; NT-proBNP, N-terminal pro-B-type natriuretic peptide; RWT, relative wall thickness; TR, tricuspid regurgitation

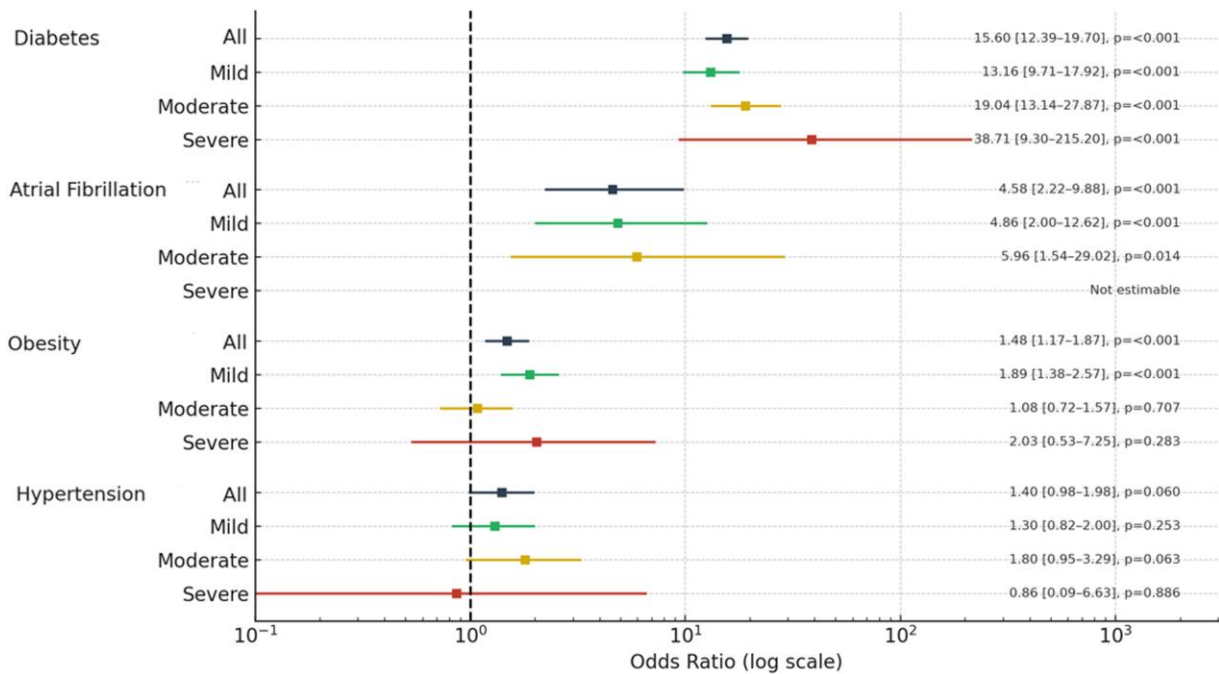


Figure 3 Association between comorbidities and heart failure with preserved ejection fraction. The forest plot displays the adjusted odds ratios (represented by filled square) with 95% confidence intervals (represented by tail) and is plotted on a logarithmic scale. Odds ratios were derived from logistic regression models and were adjusted by age and adult patients with congenital heart disease severity for the model for all adult patients with congenital heart disease. Stratified analysis by adult patients with congenital heart disease severity (mild, moderate, severe) was adjusted by age. Comorbidities tested for association with heart failure with preserved ejection fraction include diabetes, atrial fibrillation, obesity, and hypertension

Table 1 Baselinedemographic and characteristics of adult patients with congenital heart disease meeting heart failure with preserved ejection fraction criteria by adult patients with congenital heart disease severity and comparison with adult patients with congenital heart disease with normal systolic and diastolic left ventricular function

CHD classification	HFpEF by ACHD severity (n = 572)			P-value	HFpEF	Normal LV function	P-value
	Mild	Moderate	Severe				
Prevalence, n (%)	318 (13)	222 (12)	32 (18)		572 (13)	3324 (74)	
Female	106 (33)	118 (53)	22 (69)	<.0001	246 (43)	1513 (46)	<.001
Age	64 [55-73]	54 [35-66]	52 [33-62]	<.0001	60 [47-70]	41 [29-56]	<.001
Inclusion year	2023 [2022-2023]	2023 [2022-2024]	2024 [2022-2024]		2023 [2022-2024]	2023 [2022-2024]	
Comorbidities							
Atrial fibrillation	15 (4.7)	7 (3.2)	0 (0)	.1	22 (3.8)	12 (0.4)	<.001
Anaemia ^b	58 (26)	36 (27)	6 (21)	.7	100 (26)	141 (15)	<.001
Hb ^b	13.5 ± 1.7	13.5 ± 1.7	15.5 ± 3.5	.0003	13.7 ± 2.0	14.1 ± 1.5	<.001
Diabetes	168 (53)	122 (55)	12 (38)	.5	302 (53)	180 (5.4)	<.001
COPD	34 (11)	16 (7.2)	2 (6.3)	.1	52 (9.1)	66 (2.0)	<.001
KD	2 (0.6)	4 (1.8)	0 (0)	.5	6 (1.0)	1 (<0.1)	<.001
eGFR (mL/min/1.73m ³) ^c	82.3 ± 20.3	90.4 ± 26.9	77.4 ± 38.2	.2	85.0 ± 24.3	98.7 ± 19.1	<.001
Obesity ^d	76 (25)	42 (19)	6 (19)	.1	124 (22)	456 (15)	<.001
BMI (kg/m ²) ^d	26.8 ± 4.9	26.0 ± 6.0	24.9 ± 5.7	.03	26.4 ± 5.4	25.3 ± 5.0	<.001
Hypertension	31 (9.7)	17 (7.7)	2 (6.3)	.3	50 (8.7)	135 (4.1)	<.001
PAH	2 (0.6)	12 (5.4)	10 (31)	<.0001	24 (4.2)	10 (0.3)	.001
Medications							
ACE-i	82 (26)	34 (15)	1 (3.1)	.0001	117 (20)	193 (5.8)	<.001
ARB	108 (34)	54 (15)	4 (13)	.001	166 (29)	373 (11)	<.001
BB	173 (54)	100 (45)	16 (50)	.08	289 (51)	497 (15)	<.001
MRA	58 (18)	53 (24)	10 (31)	.03	121 (21)	29 (0.9)	<.001
SGLT2i	21 (6.6)	11 (5.0)	1 (3.1)	.3	33 (5.8)	27 (0.8)	<.001
Cardiac electronic devices							
PM	21 (6.6)	11 (5.0)	1 (3.1)	.3	33 (5.8)	60 (1.8)	<.001
ICD	5 (1.6)	13 (5.9)	1 (3.1)	.03	19 (3.3)	22 (0.7)	<.001
TAPSE ^a	2.1 ± 0.6	1.8 ± 0.5	1.7 ± 0.5	<.0001	2.0 ± 0.6	2.2 ± 0.6	<.001
Normal (≥1.7)	172 (76)	100 (59)	12 (48)		284 (67)	1934 (87)	
Reduced (<1.7)	54 (24)	70 (41)	12 (48)		137 (33)	285 (13)	

Continued

Table 1 Continued

CHD classification	HFpEF by ACHD severity (n = 572)			P-value	HFpEF	Normal LV function	P-value
	Mild	Moderate	Severe				
HF diagnosis	4 (1.3)	14 (6.3)	5 (16)	<.0001			
Hospitalization rate (/100 PY)	30.4 [27.8–33.2]	32.4 [29.2–35.9]	49.5 [39.6–61.9]	.0004	32.2 [30.2–34.3]	5.6 [5.3–6.0]	<.0001
Sensitivity analysis ^e	21.1 [17.5–25.5]	28.4 [22.8–35.4]	68.4 [39.7–117.8]	.0003	24.7 [21.5–28.4]	5.5 [4.9–6.1]	<.0001

Data presented as number (%), median [IQR], or mean ± SD. Hospitalization rates are model-based estimates from Poisson regression, numbers in [] represent 95% confidence interval. ACHD, adult congenital heart disease; HFpEF, heart failure with preserved ejection fraction; COPD, chronic obstructive lung disease; eGFR, estimated glomerular filtration rate; BMI, body mass index; ACE-i, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BB, beta blocker; MRSA, mineralocorticoid receptor antagonist; SGLT2i, sodium-glucose cotransporter-2 inhibitor; PM, pacemaker; ICD, implantable cardioverter-defibrillator.

^a151 (26.4%) in HFpEF, 1105 (33.2%) in normal LV function missing data.

^b190 (33.2%) in HFpEF/1992 (59.9%) in normal LV function missing data.

^c200 (35%) in HFpEF/1983 (59.7%) in normal LV function missing data.

^d11 (1.9%) in HFpEF/267 (8%) in normal LV function missing data.

^eHospitalization rate was calculated based on hospitalizations that occurred from day of echocardiography to end of follow-up (either death or 1 September 2024).

patients compared with patients with normal LV function [adjusted IRR: 3.4 (95% CI 2.8–4.1), *P* < .0001].

Heart failure diagnosis and medication

Among patients meeting HFpEF criteria, only 4.0% had a recorded HF diagnosis and only 5.8% received a SGLT2 inhibitor, with no significant trend across ACHD severities (Table 1). Pharmacotherapy across LV function groups is shown in Supplementary data online, Figure S4.

Subanalysis of patients with systemic right ventricle and univentricular physiology

The subanalysis included 117 patients with systemic RV and 131 with a univentricular heart. Among the univentricular patients, the systemic ventricle was morphologically right in 49 (37%) and left in 82 (63%) patients. Among the 248 patients, the systemic ventricular ejection fraction was preserved in 58% (*n* = 143), mildly reduced in 22% (*n* = 54), and reduced in 20% (*n* = 51). Within the preserved ejection fraction group, 27% (*n* = 38) received diuretics. In total, 2.8% (*n* = 4) of patients with preserved ejection fraction fulfilled the ESC HFpEF criteria. In comparison, elevated NT-proBNP was observed in 5.6% (*n* = 3) and 23.5% (*n* = 12) of patients with mildly reduced and reduced ejection fraction, respectively. Additional demographics are presented in Table 2.

Discussion

To our knowledge, this is the first study to assess the prevalence of HFpEF in a large cohort of patients with ACHD. We found that a markedly higher proportion of ACHD patients met the ESC criteria for HFpEF compared with prevalence estimates reported in the general population of similar age.^{1,19} Meeting the HFpEF criteria was particularly common among the older patients within the ACHD population and was associated with a higher burden of cardiovascular risk factors and hospital admissions. Despite these findings, only a minority had a documented HF diagnosis or received guideline-directed treatment with SGLT2 inhibitors (Structured Graphical Abstract).

The prevalence of HFpEF in the general population is reported to be between 1.0% and 5.5%, although it is believed to be underdiagnosed and expected to continue rising in the coming years.^{2,3,9,20,21} In our ACHD cohort, the prevalence was markedly higher.

Key contributors to the high prevalence of HFpEF in ACHD are assumed to be structural abnormalities, altered haemodynamic, previous cardiac surgeries or interventions, and early-life cyanosis.^{3,4,22} The higher prevalence of HFpEF among patients with complex conditions such as TOF and Eisenmenger syndrome is therefore perhaps not surprising. However, the prevalence was also high among patients with mild ACHD, which may not be fully explained by the congenital heart defects alone, considering the higher prevalence of risk factors among this subpopulation.

Patients meeting the HFpEF criteria had a much higher prevalence of known risk factors such as diabetes, hypertension, obesity, and AF compared with ACHD patients with normal LV function. Previous studies have also demonstrated that ACHD patients generally carry a higher burden of cardiovascular

Table 2 Baseline characteristics of adult patients with congenital heart disease with a systemic right ventricle or a univentricular heart

	Systemic RV (n = 117)	Univentricular (n = 131)
Female	47 (40)	49 (37)
Age	40 [35–49]	28 [23–34]
EF class		
Preserved	55 (47)	88 (67)
Mildly reduced	23 (20)	31 (24)
Reduced	39 (33)	12 (9.2)
Comorbidities		
Atrial Fibrillation	114 (97)	129 (98)
Anaemia ^a	5 (13)	15 (11)
Hb ^a	14.7 ± 1.7	16.0 ± 2.1
Diabetes	16 (14)	10 (7.6)
COPD	1 (0.9)	2 (1.5)
CKD	1 (0.9)	1 (0.8)
eGFR (mL/min/ 1.73m ²) ^b	97 ± 21	108 ± 17
Obesity ^c	19 (17)	12 (9.3)
BMI (kg/m ²) ^c	25.9 ± 4.6	23.4 ± 4.4
Hypertension	18 (15)	10 (7.6)
PAH	2 (1.7)	15 (11)
Medications		
ACE-i	25 (21)	25 (19)
ARB	13 (11)	6 (4.6)
BB	48 (41)	36 (27)
MRA	19 (16)	27 (21)
SGLT2i	7 (6.0)	3 (2.3)
Cardiac electronic devices		
PM	19 (16)	0 (0)
ICD	13 (11)	0 (0)

Data presented as number (%), median [IQR], or mean ± SD.

ACHD, adult congenital heart disease; HFpEF, heart failure with preserved ejection fraction; Hb, haemoglobin; COPD, chronic obstructive lung disease; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; BMI, body mass index; ACE-i, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BB, beta blocker; MRA, mineralocorticoid receptor antagonist; SGLT2i, sodium-glucose cotransporter-2 inhibitor; PM, pacemaker; ICD, implantable cardioverter defibrillator.

^a71 (28.6%) missing data.

^b72 (29%) missing data.

^c9 (3.6%) missing data.

in this population.⁵ Notably, several of these risk factors, particularly hypertension, diabetes, and obesity, are known contributors to the development of HFpEF. While not all risk factors are modifiable, those that are may not be adequately addressed by the congenital specialists, as reported by Bauer *et al.*²³ If the development of HFpEF in patients with mild ACHD is indeed driven largely by these risk factors, awareness and management within ACHD care are essential.

The combination of structural cardiac abnormalities and modifiable HFpEF risk factors could probably explain why all ACHD severity groups had a high prevalence of HFpEF, although they differed in the distribution of age and contributing HFpEF risk factors.

Importantly, our classification of HFpEF was based on the 2021 ESC HF guidelines, which were developed for the general population and have not been validated in ACHD. Patients with right-sided volume or pressure overload (e.g. ASD, TOF) may receive diuretics or have elevated tricuspid regurgitation velocity due to RV dysfunction, thus carrying the risk of being misclassified as HFpEF. Although our classification was based primarily on left-sided echocardiographic parameters which reflects underlying LV pathology, contribution from right-sided ventricular dysfunction may coexist. In conjunction with this, we evaluated RV systolic function using TAPSE. We found that among those meeting the HFpEF criteria with available measurements (*n* = 421), 67% had a preserved TAPSE (≥ 17 mm). Nevertheless, the proportion of right HF and possible ventricular interdependence in the identified HFpEF patients is not negligible. As the therapeutic approach may differ in these patients, our study reminds the importance of an ACHD-specific criteria for HFpEF diagnosis, alongside our objective to increase the awareness of the diagnosis among ACHD patients. Furthermore, some echocardiographic parameters such as E/e' ratios may have reduced accuracy in ACHD patients with abnormal septal motion or prior septal surgery. In addition, NT-proBNP, while a well-established marker for elevated filling pressures, can also be influenced by other cardiac or non-cardiac comorbidities, e.g. PAH and CKD. Nonetheless, in our cohort, only four patients (2%) were identified solely on NT-proBNP levels, and PAH was present in 49 patients (1%) and CKD in 9 patients (0.1%), suggesting a limited contribution of these confounders in our identified HFpEF patients.

Our findings are consistent with previous studies by Savarese *et al.*²⁰ and Dunlay *et al.*¹¹ showing that, similar to HF with reduced ejection fraction (HFrEF), HFpEF is associated with adverse outcomes. Patients meeting the HFpEF criteria experienced a near five-fold higher rate of cardiovascular hospitalizations over a 5-year period compared with those with normal LV function. While a high rate of hospitalization might be expected in patients with complex ACHD, we also observed a substantial hospitalization burden in those with mild and moderate ACHD. This finding is supported by a recent Danish registry study by Lim *et al.*, which reported rising cardiovascular hospitalization rates over time in patients with mild ACHD.²⁴

Despite increased cardiovascular hospitalizations, only 4.0% of ACHD patients meeting the HFpEF criteria had a documented HF diagnosis. The recent introduction of a specific ICD-10 code for HFpEF in Denmark may partly explain this, as might the tendency to attribute symptoms such as dyspnoea and

risk factors than the general population.^{5–8,10} Reflecting this concern, the 2023 ESC Working Group on ACHD issued a call to action to prevent early-onset acquired cardiovascular disease

impaired exercise tolerance to the underlying congenital defect²⁵ rather than to diastolic dysfunction.

In recent years, large trials such as DELIVER¹² and EMPEROR-Preserved¹³ have shown that SGLT2 inhibitors reduce hospitalization risk in patients with HFpEF. Accordingly, these agents have been incorporated into the 2023 updated ESC HF guidelines as a class IA recommendation.¹⁶ However, patients with ACHD have generally been excluded from these trials, resulting in limited evidence to guide medical therapy in this population. Recent studies^{26–28} suggest that SGLT2 inhibitors are well tolerated and may improve symptoms and reduce HF hospitalizations in ACHD-HF patients, including rare HFpEF cases. In our cohort, SGLT2 inhibitor use in HFpEF was only 5.8% despite a high diabetes prevalence. Likely reasons include under-recognition of HFpEF, the recent introduction of guideline recommendations, and, in some cases, concerns about renal function or blood pressure, although these factors probably do not explain the majority of underuse, as most patients in our study had preserved renal function.

Patients with a systemic RV or a univentricular heart present additional diagnostic challenges. In our subpopulation, only a small proportion of patients met the ESC HFpEF criteria, yet many are likely to have unrecognized diastolic impairment masked by coexisting systolic dysfunction. Chronic volume or pressure overload, multiple surgeries, and early-life cyanosis likely contribute to a complex HF phenotype characterized by overlapping systolic and diastolic dysfunction. Conventional left-sided HFpEF definitions have limited sensitivity in these patients, underscoring the need for ACHD-specific diagnostic criteria that capture the full spectrum of ventricular dysfunction.

Strengths and limitations

A major strength of this study is its large, well-characterized, multicentre cohort of adult patients with congenital heart disease. Patients were included from both urban and rural settings, spanning high- and low-income areas, and representing the full spectrum of ACHD severity. In Denmark, ACHD care is centralized within the public healthcare system, ensuring equal access for all citizens across the entire region regardless of socioeconomic status. Diagnoses were based on systematically documented electronic health records and were validated by an ACHD specialist. However, several limitations should be acknowledged.

The ESC HFpEF criteria have not been validated in ACHD and may have variable specificity in patients with complex anatomy or right-sided lesions, as discussed above. Some echocardiographic and biomarker measurements may be less reliable in this population; nonetheless, they currently represent the best available tools, as no ACHD-specific diagnostic guidelines for HFpEF exist. The use of diuretic therapy as a surrogate marker for symptomatic HFpEF has limitations, as it may reflect volume overload but does not necessarily capture symptoms such as dyspnoea or reduced exercise capacity. Consequently, some patients with HFpEF may have been missed. However, 4.8% of the patients were classified as having HFpEF based solely on mineralocorticoid receptor antagonist treatment, which in some cases may have been prescribed for indications other than HF. In addition, due to the small sample size of patients with severe ACHD, findings in this subgroup should be interpreted with

caution. In addition, as blood investigation is not routinely performed at each ACHD follow-up visit, there could be a lapse of time between the most recent echocardiography and blood investigation, as well as a widespread interval between the three most recent blood results. However, the impact of this limitation is thought to be minimal, as our data shows that 99% of the patients who fulfilled the HFpEF diagnostic criteria were diagnosed based on echocardiographic parameters and the median interval between the earliest and latest date of the three most recent blood investigations is 43 days (IQR 0–364). Furthermore, we acknowledge that some variables, such as haemoglobin, eGFR, and TAPSE, had higher proportions of missing data. However, these parameters were not included as variables in regression analyses but are presented descriptively to provide a more complete clinical characterization of the cohort. We therefore believe that retaining the patients with missing data for these variables in the study cohort does not affect the validity of the analyses.

Furthermore, the use of registry-based data may introduce limitations related to the validity of the comorbidity diagnoses, which we have incorporated information on prescribed medication to support and validate these diagnoses. Lastly, we do not have emigration data to account for loss to follow-up. However, because transfer of ACHD care to other regions of Denmark would require agreement from the responsible ACHD cardiologist, it is from our experience that mobilization of ACHD patients across the Danish regions or across countries is a very rare occasion. Having said so, we acknowledge that a small number of hospitalizations may have occurred outside our regions, but this only happen occasionally and often the hospitalization is not cardiovascular related.

Conclusion

Adults with congenital heart disease have a higher prevalence of patients fulfilling criteria for HFpEF compared with reports from the general population. Heart failure with preserved ejection fraction was associated with older age, a greater burden of cardiovascular risk factors, and increased hospitalization rate. Yet, only a few patients had a recorded HF diagnosis or received guideline-recommended treatment with SGLT2 inhibitors. These findings highlight the need for clinical awareness of HFpEF across the entire ACHD spectrum to enable timely diagnosis, prevention, and management of HFpEF in this population.

Supplementary data

Supplementary data are available at *European Heart Journal* online.

Declarations

Disclosure of Interest

L.K. reports speaker's honoraria from AstraZeneca, Boehringer Ingelheim, Novartis, and Novo. S.S.S. reports travel and congress fee support from Novartis for attendance at the ESC

Heart Failure Congress 2025. The remaining authors have no conflicts of interest to declare.

Data Availability

The data underlying this article is from the medical electronic patient files. A permission was granted from the institutional review boards (p-2024-17036).

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Nothing to declare.

Ethical Approval

The institutional review boards approved the study (p-2024-17036), which adheres to the ethical principles of the Declaration of Helsinki and is consistent with Good Clinical Practice and applicable regulatory requirements.

Pre-registered Clinical Trial Number

Not applicable.

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